CHAPTER ONE

INTRODUCTION

Ordinal logistic regression is a statistical modeling technique used to analyze and predict relationships between independent variables and an ordinal (ranked) dependent variable. It is commonly employed in various fields where the outcome of interest is ordered or ranked, such as psychology, sociology, education, and healthcare. Its versatility in handling ordered outcomes makes it a valuable tool for understanding and predicting various phenomena across different fields. On this paper we will use it to study depression levels among cancer care givers.

1.1 Background Information

Depression is a mental health disorder characterized by persistent feelings of sadness, loss of interest or pleasure in activities, changes in appetite or sleep patterns, fatigue, difficulty concentrating, and thoughts of self-harm or suicide. It is a complex condition that can significantly impact individuals' emotional well-being, cognitive functioning, and overall quality of life.

Depressed individuals may have difficulty concentrating and making decisions, which can hinder their productivity in various aspects of life. Rehabilitation centers for depression have been established to provide specialized treatment facilities that provide comprehensive care for individuals experiencing moderate to severe depression.

1.2 Statement of the Problem

Researchers have been faced with the problem of identifying potential predictors of depression among cancer caregivers. The problem is to identify and to quantify the important variables linked to depression. In order to better understand the complex processes causing caregiver depression and to guide the creation of effective therapies and support systems, we have identified and quantified the important variables linked to depression and developed Ordinal logistic regression model to understand the relative influence of these predictors.

1.3 Objective of the Study

1.3.1 General objectives

To model depression severity among caregivers in a rural setting using ordinal logistic regression model.

1.3.2 Specific objectives

1. To identify the significant predictors that contribute to depression among cancer caregivers in a rural setting
2. To assess the performance of the ordinal logistic regression model in predicting depression among cancer caregivers
3. To provide evidence-based recommendations for interventions and support systems aimed at addressing depression among cancer caregivers

1.4 Methodology framework

A cross sectional research design was used in a rural setting among cancer caregivers on 367 participants. The independent variables were the age of patient and participant and the level of income per month in a household. These were used to determine their relationship with the response variable which in this case is depression. A PHQ-9 scale was used to asses depression and the identified predictors and was administered using a face-to-face interview to gather data from the identified caregivers. Privacy and confidentiality of the data was ensured. The data was cleaned and outliers checked. An ordinal logistic regression was performed using R software to explore the relationship between the variables of interest. The results were interpreted and goodness of fit evaluated. A discussion of the findings took place, conclusions were drawn and recommendations made accordingly.

1.5 Justification of the Study

Cancer caregiver depression is a serious mental health issue that can have a negative impact on both the caregivers and the people they are caring for. Ordinal logistic regression analysis used to examine the depression predictors in the rural context can assist identify the unique elements leading to depression in these environments and direct the creation of specialized remedies. The results of this study can help with the creation and enhancement of rural cancer caregiver support systems. Healthcare professionals, policymakers, and support groups can create targeted treatments that meet the unique needs and difficulties of caregivers in these circumstances by recognizing the predictors of depression. Access to mental health treatments, strengthening social support systems, and putting caregiver relief measures into place.

CHAPTER 2

LITERATURE REVIEW

In this chapter, a review of the existing research and scholarship related to mental health and its effects on a person's productivity as well as the interaction between an individual's cognitive, affective and somatic symptoms of depression is made. Key insights from previous research have immensely contributed to recent advances in mental health understanding.The following are some of the previous works.

A research study conducted by Martin et al. (2009), investigated whether different types of health promotion interventions in the workplace reduced depression and anxiety symptoms. A systematic review and meta-analysis of the literature was undertaken on workplace health promotion published during the period 1997-2007.Studies were considered eligible for inclusion if they evaluated the impact of an intervention using a valid indicator or specific measure of depression or anxiety symptoms. The standardized mean difference was calculated for each of the following three types of outcome measures; depression, anxiety and composite mental health. Altogether,22 studies met the inclusion criteria, with a total sample size of 3409 employees post intervention, and 17 of the studies were included in the meta-analysis representing 20 interventions-control comparisons. The pooled results indicated small but positive overall effects of the interventions with respect to symptoms of depression and anxiety, but no effect on composite mental health measures. The interventions that included a direct focus on mental health had a comparable effect on depression and anxiety symptoms as did the interventions with an indirect focus on risk factors.

Jain et al. (2013) examined the burden of depression on work productivity in the US workforce. The Patient Health Questionnaire for depression severity and the Health and Work Performance Questionnaire and Work Productivity and Activity Impairment (WPAI) Questionnaire for absenteeism and presenteeism were used to survey 1051 full-time employees in February 2010. The results indicated that out of the 1051 employees with depression, 40.3% had no depressive symptoms at the time of the survey,30.4% had mild depression, 15.8% had moderate depression,7.8% had moderately severe depression, and 5.8% had severe depression. All levels of depression were associated with decreased work productivity as severity increased, presenteeism and absenteeism worsened.

In another study Serpa et al. (2014), investigated the major problems among Veterans specifically anxiety, depression and suicidal ideation. Pre- and Post Mindfulness Based Stress Reduction (MBSR) questionnaires were used to collect data on MBSR's effectiveness among 79 veterans at an urban Veterans Health Administration medical facility and were used for comparison among individuals. A meditation analysis was conducted to determine whether changes in mindfulness were related to changes in the other outcomes for a period of 9 weekly sessions. The results indicated significant reductions in anxiety, depressions, and suicidal ideation after MBSR training as well as improved mental health functioning scores although pain intensity and physical health functionality did not register improvements.

Laing and Jones (2016) investigated the mediation effect of anxiety and depression on the relationship between perceived health-promoting workplace culture and presenteeism. Paper surveys were distributed to 4703 state employees. The variables that were investigated included symptoms of depression, anxiety, perceived workplace support for healthy living inclusive of physical activity and presenteeism. Correlational analysis was used to asses the relationships among culture, mental health and productivity. The results indicated that indirect effects of workplace culture on productivity, mediated by anxiety and depression symptoms were significant. Healthy living culture and anxiety were significantly negatively associated and anxiety and presenteeism were significantly positively associated.

Another study by Magtibay et al. (2017) assessed the efficacy of blended learning which was intended to decrease stress and burnout among nurses through the use of the Stress Management and Resiliency Training (SMART) program. Consistent with blended learning, participants chose the format that met their learning styles and goals; Web-based, independent reading or facilitated discussions. The end points of mindfulness, resilience, anxiety, stress, happiness, and burnout were measured at baseline, postintervention, and 3-month follow-up to examine within-group differences. The results indicated statistically significant, clinically meaningful decreases in anxiety, stress, and burnout and increases in resilience, happiness, and mindfulness.

Levis et al. (2019), evaluate the accuracy of the Patient Health Questionnaire-9 (PHQ-9) for screening major depression in adults using individual participant data meta-analysis. The study, which was conducted by the DEPRESsion Screening Data (DEPRESSD) Collaboration, analyzed data from 58 studies, with a total of 17 357 participants. It was found that combined sensitivity and specificity was maximized at a cut-off score of 10 or above among studies using a semi-structured interview.

Secondly across cut-off scores of 5-15, sensitivity with semi-structured interviews was 5%-22% higher than for fully structured interviews.

Leung et al. (2020) investigated the factor structure, reliability, and construct validity of the Patient Health Questionnaire-9 (PHQ-9) in a community sample of 10 933 Chinese adolescents. Confirmatory factor analysis (CFA) was used to examine the one-factor structure of the PHQ-9, and multiple-group CFAs were performed to test for measurement invariance across gender and age subgroups. Internal consistency reliability was evaluated using Cronbach alpha, and construct validity was assessed using Pearson correlation coefficients between PHQ-9 scores and anxiety, self-esteem and perceived control. The results indicated that a one-factor model with three pairs of items correlations fitted the PHQ-9 data well and that measurement invariances by age and gender were supported. The PHQ-9 was also strongly correlated with anxiety, self-esteem and perceived control.

In a recent study, Levis et al. (2020), a meta-analysis of 44 studies was conducted to compare the sensitivity and specificity of the PHQ-2 alone and combined with PHQ-9.The Individual participant data were obtained from 100 eligible studies comprising of 44 318 participants, of which 4572 had major depression. The authors found that the combination of PHQ-2 followed by PHQ-9 had similar sensitivity but higher specificity compared with PHQ-9.

In a more recent study, Lister et al. (2023) conducted a qualitative study that investigated the barriers and enablers to mental well being and academic success that students experienced in distance learning. Narrative inquiry was used whereby students shared their own stories and tutors shared stories about the students they had supported. In total, 60 themes were identified, consisting of 155 sub-themes and 783 coded references. The barriers and enablers were then clustered into three overall categories; study-related barriers/enablers, skills-related barriers/enablers and environmental barriers/enablers, and 10 sub-categories, curriculum, tuition, assessment, social skills, self-management, study skills, spaces, systems, people and life. It was found that barriers and enablers were existent in the majority of themes, implying that different aspects of study could be both barriers or enablers for mental well being and study success, depending on who was experiencing them and how, for example, assessment feedback was a barrier for students when it was negative, but was an enabler when it was constructive and took place in an environment of trust.

In summary, the past research works on mental health have been conducted using

Meta-analysis, Qualitative techniques, CFA and Narrative inquiry. ordinal logistic regression will be conducted to examine the relationships between the independent variables (age of patient, age of participant, marital status, education, social support ,and level of income) and the dependent variable (level of depression). The findings of this research will contribute to our understanding of the factors influencing depression among caregivers in rural settings and may inform the development of targeted interventions to support these individuals.

CHAPTER THREE

Ordinal Logistic Regression

3.1 Introduction

This chapter presents the the theory of the ordinal logistic regression model.

3.2 Ordinal Logistic Regression

Ordinal logistic regression is a statistical method utilized to analyze the association between a categorical with natural ordering and one or more explanatory variableS. The explanatory variables can be either continuous or categorical in nature. Unlike logistic regression, which deals with binary responses, ordinal logistic regression extends this concept to handle responses with multiple ordered levels (Agresti, 2019).

3.3 Assumptions of the model:

For the results of ordinal logistic to hold, certain assumptions have to be met. The assumptions associated with Ordinal Logistic Regression are:

- Proportional Odds Assumption:
  The proportional odds assumption states that the relationship between the independent variables and the cumulative odds of being in a higher category remains constant across all levels of the dependent variable. This assumption implies that the coefficients associated with the independent variables are the same across different levels of the dependent variable.

- Independence of Observations:
  Each observation or data point is independent of others. This assumption requires that the observations are not influenced by each other, meaning that the outcome of one observation does not affect the outcome of another.

- Ordinality of the Dependent Variable:
  Ordinal logistic regression assumes that the dependent variable is measured on an ordinal scale. This means that the categories of the dependent variable have a natural order and the distance between categories is not necessarily equal.

- Absence of Perfect Separation:
  Perfect separation occurs when there is a combination of independent variable

values that perfectly predicts the outcome category. In such cases, the model cannot be estimated or interpreted properly.

- Adequate Sample Size:
  Having an adequate sample size is important for reliable results. The sample size should be large enough to ensure stable parameter estimates and valid statistical inference. Small sample sizes can lead to imprecise estimates and low statistical power.

3.4 The Model

Let **Y** be an ordinal outcome with **j** categories, $j \geq 2$.
Suppose there are p covariates $X\_1, X\_2, ..., X\_p$
Then $P(Y \leq j)$ is the cumulative probability of **Y** less than or equal to a specific category **j=1,...,j-1**.
The odds of being less than or equal a particular category can be defined as $\frac{P(Y \leq j)}{P(Y > j)}$

The ordinal logistic regression model is defined as follows:

$$\text{logit}(P(Y \leq j)) = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p \tag{3.1}$$

There exists three parameterizations of the ordinal logistic regression model

1. model 1:
$$log\frac{P(Y \leq j)}{P(Y > j)} = \alpha_j - \beta_1 X_1 - \beta_2 X_2 - \ldots - \beta_p X_p \tag{3.2}$$

2. model 2:
$$log\frac{P(Y \leq j)}{P(Y > j)} = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p \tag{3.3}$$

3. model 3:
$$log\frac{P(Y \leq j)}{P(Y \leq j)} = \alpha_j - \beta_1 X_1 - \beta_2 X_2 - \ldots - \beta_p X_p \tag{3.4}$$

where:

- $P(Y \leq j)$ is the cumulative probability of the response variable Y being in category j or lower,

- α_j is the intercept parameter specific to category j,
- $\beta_1, \beta_2, \ldots, \beta_p$ are the coefficients associated with the independent variables $X_1, X_2, \ldots, X_p$, respectively.

In the re-parametrized model 1, a negative sign is incorporated in the coefficients to establish a direct correspondence between the slope and the ranking. Therefore, a positive coefficient in the model indicates that as the value of the explanatory variable increases, the likelihood of a higher ranking also increases. This re-parametrization helps in interpreting the coefficients in a more intuitive manner, as a positive coefficient signifies a positive effect on the response variable, aligning with our expectation that higher values of the explanatory variable lead to higher rankings.

3.5 Interpretation of model Coefficients

- The intercept parameter α_j represents the log-odds of the response variable being in category j or lower when all independent variables are zero.
- The coefficient β_k associated with independent variable X_k represents the change in the log-odds of the response variable for a one-unit increase in X_k, while holding all other variables constant.

To obtain the odds ratio, exponentiate the coefficient:

$$\text{Odds Ratio} = \exp(\beta_k)$$

The odds ratio indicates how the odds of being in a higher category change with a one-unit increase in the corresponding independent variable, assuming the other variables are held constant.

3.6 Prediction of New Observations

The predicted probabilities and proportional odds can be obtained for ordinal logistic regression. Using model 2,

$$log\frac{P(Y \leq j)}{P(Y > j)} = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p \tag{3.5}$$

the predicted probabilities can be calculated as

$$log\frac{P(Y \leq j}{P(Y > j)} = \frac{e^{\alpha_j+\beta_1 x_1+\beta_2 x_2+\ldots+\beta_p x_p}}{1+e^{\alpha_j+\beta_1 x_1+\beta_2 x_2+\ldots+\beta_p x_p}} Predicted probabilities \tag{3.6}$$

When the assumption of proportional odds holds, the predicted probabilities derived from the model will closely align with the observed proportions.

3.7 Estimation of the Model Parameters

$$E[Y_i] = \pi_i \tag{3.7}$$

and

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip} = \mathbf{x}_i^T \boldsymbol{\beta}, \tag{3.8}$$

where $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})$ prime and $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_p)$prime . Therefore,

$$E[Y_i] = \frac{e^{\mathbf{x}_i^T \boldsymbol{\beta}}}{1 + e^{\mathbf{x}_i^T \boldsymbol{\beta}}}$$

The likelihood for n observations then is;

$$L = \prod_{i=1}^{n} \frac{\exp(\acute{x_i}\beta)}{1 + \exp(x_{\acute{x_i}\beta)} \left(\prod_{i=1}^{n}(1 - \exp(\acute{x_i}\beta))\right)^{n-\sum_{i=1}^{n} Y_i}}, (3.9)$$

and the log-likelihood is

$$l = \sum_{i=1}^{n} Y_i \log\left(\frac{\exp(\acute{x_i}\beta)}{1 + \exp(\acute{x_i}\beta)}\right) + (1 - Y_i)\log\left(1 - \frac{\exp(\acute{x_i}\beta)}{1 + \exp(\acute{x_i}\beta)}\right). \tag{3.10}$$

3.8 Confidence intervals for the coefficients and the odds ratios

In the model

$$log\frac{P(Y \leq j}{P(Y > j)} = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p$$

a $(1 - \alpha/2) \times 100\%$ confidence interval for β_j, where $j = 1, \ldots, p$, can easily be calculated as

$$\hat{\beta}_j \pm Z_{1-\alpha/2} \cdot \text{se}(\hat{\beta}_j).$$

The $(1-\alpha/2)\times 100\%$ confidence interval for the odds ratio over a one unit change in x_j is given by

$$\left(\exp(\hat{\beta}_j - Z_{1-\alpha/2}\cdot \mathrm{se}(\hat{\beta}_j)), \exp(\hat{\beta}_j + Z_{1-\alpha/2}\cdot \mathrm{se}(\hat{\beta}_j))\right).$$

3.9 Hypothesis testing Likelihood Ratio Test

The likelihood ratio test (LRT) statistic calculates the ratio between the likelihood at the hypothesized parameter values and the likelihood of the data at the maximum likelihood estimate(s). This statistic is derived from the following formula: For ordinal logistic regression, the likelihood ratio test (LRT) statistic can be calculated as:

$$LR = -2\log\left(\frac{L(\text{at H0})}{L(\text{at MLE(s)})}\right) = -2l(\text{H0}) + 2l(\text{MLE}) \tag{3.11}$$

For large n, the LRT statistic follows a chi-square distribution with degrees of freedom equal to the number of parameters being estimated.

For a given binary outcome, if the hypothesis is $H0: p = p_0$ versus $HA: p \neq p_0$, the log-likelihoods are:

$$l(\text{H0}) = \frac{1}{n}\sum_{i=1}^{n} Y_i \log(p_0) + \left(1 - \frac{1}{n}\sum_{i=1}^{n} Y_i\right)\log(1-p_0)$$
$$l(\text{MLE}) = \frac{1}{n}\sum_{i=1}^{n} Y_i \log(\hat{p}) + \left(1 - \frac{1}{n}\sum_{i=1}^{n} Y_i\right)\log(1-\hat{p})$$

where Y_i represents the binary outcome variable, p_0 is the hypothesized probability, and $\hat{p}$ is the maximum likelihood estimate for the probability.

The LRT statistic is given by:

$$LR = -2\left(\frac{1}{n}\sum_{i=1}^{n} Y_i \log\left(\frac{p_0}{\hat{p}}\right) + \left(1 - \frac{1}{n}\sum_{i=1}^{n} Y_i\right)\log((1-p_0)(1-\hat{p}))\right) \tag{3.12}$$

where $LR \sim \chi^2_1$.

CHAPTER FOUR

DATA ANALYSIS

4.1 Introduction

In this chapter, we discuss the sources of our data and the process of the preparation. We also explore the characteristics of data and finally we analyze and interpret the results.

4.2 Data for the Study

The data utilized in this study is Depression among cancer caregivers in a rural setting dataset published on figshare repository(Kagawa et al., 2021). Trained research assistants (RA) with training in counseling and Responsible Conducted of Research (RCR), administered the translated 30-minute-long questionnaire (written form). This questionnaire captured the participants' sociodemographic characteristics; age, sex, marital status, level of education, employment status, religion, and monthly income. Depression was assessed using the patient health questionnaire. All participants provided written consent to participate in the study. The sample size of 700 was calculated using the Kish-Leslie formula, basing on an estimated prevalence of 38.2% among caregivers of cancer patients, level of attrition of 10%, 95% confidence interval and margin of error 5%.
Caregivers of cancer patients to participate in the study by convenience sampling based on the caregivers present during the time of data collection - in the evening (convenience time for most caregivers) when the caregivers were not so occupied with caregiving activities. Caregivers who were already receiving mental health care for depression (as reported by the caregiver) were excluded from the study. Because the study was screening for individuals who need treatment and those were already on treatment.
In this study, we utilized the "Age of participant" variable which represents the age of the caregiver who responded to the questionnaire, "Age of patient" variable, "Household income" which represents the amount of money the household earns on a monthly basis. The dependent variable (depression) was measured on ordered four levels; None=1,mild=2, moderate=3, severe=4 and profound=5.

4.3 Data Preparation

```r
str(dataset)
```

```
## 'data.frame':	700 obs. of  9 variables:
##  $ age_participant: num  54 34 44 46 44 39 55 39 60 39 ...
##  $ age_patient    : num  51 53 25 38 25 32 21 23 58 40 ...
##  $ sex_participant: int  1 1 0 1 0 0 0 0 0 0 ...
##  $ sex_patient    : int  0 1 0 1 1 0 1 0 0 1 ...
##  $ marital_status : chr  "single" "single" "divorced" "married" ...
##  $ education      : chr  "secondary" "tertiary" "secondary" "primary" ...
##  $ income_ksh     : num  69596 182616 42987 1748831 13415 ...
##  $ social_support : int  0 1 0 0 0 1 0 0 0 0 ...
##  $ depression     : Factor w/ 5 levels "1","2","3","4",..: 2 1 2 2 1 2 2 1 1 5 ...
```

```r
head(dataset, 10)
```

```
##    age_participant age_patient sex_participant sex_patient marital_status
## 1               54          51               1           0         single
## 2               34          53               1           1         single
## 3               44          25               0           0       divorced
## 4               46          38               1           1        married
## 5               44          25               0           1        married
## 6               39          32               0           0         single
## 7               55          21               0           1         single
## 8               39          23               0           0       divorced
## 9               60          58               0           0         single
## 10              39          40               0           1       divorced
##    education income_ksh social_support depression
## 1  secondary      69596              0          2
## 2   tertiary     182616              1          1
## 3  secondary      42987              0          2
## 4    primary    1748831              0          2
## 5  secondary      13415              0          1
## 6   tertiary    1474274              1          2
## 7  secondary     288196              0          2
## 8   tertiary      90785              0          1
## 9    primary     214242              0          1
## 10   primary     155082              0          5
```

```r
View(dataset)
```

```r
#data set preparation

# education levels
education_mapping <- c("primary" = 1, "secondary" = 2, "tertiary" = 3)

# marital status levels
marital_status_mapping <- c("married" = 1, "single" = 2, "divorced" = 3)

# Mutate the 'education' and 'marital_status' columns with numeric values and also
dataset <- dataset %>%
  mutate(education = education_mapping[education],
         marital_status = marital_status_mapping[marital_status],
         income_ksh=income_ksh/100000)

View(dataset)
df<-as.data.frame(dataset)
class(df)

## [1] "data.frame"

names(df)

## [1] "age_participant" "age_patient"     "sex_participant" "sex_patient"
## [5] "marital_status"  "education"       "income_ksh"      "social_support"
## [9] "depression"
```

4.4 Exploratory Data Analysis

The data set was subjected to exploratory data analysis with the objective to gain a better understanding of it. All the analysis in this study were done on R software.

```r
# Summary statistics
summary(df)

##  age_participant  age_patient     sex_participant   sex_patient
##  Min.   :10.00   Min.   :-22.00   Min.   :0.0000   Min.   :0.00
##  1st Qu.:33.00   1st Qu.: 31.00   1st Qu.:0.0000   1st Qu.:0.00
```

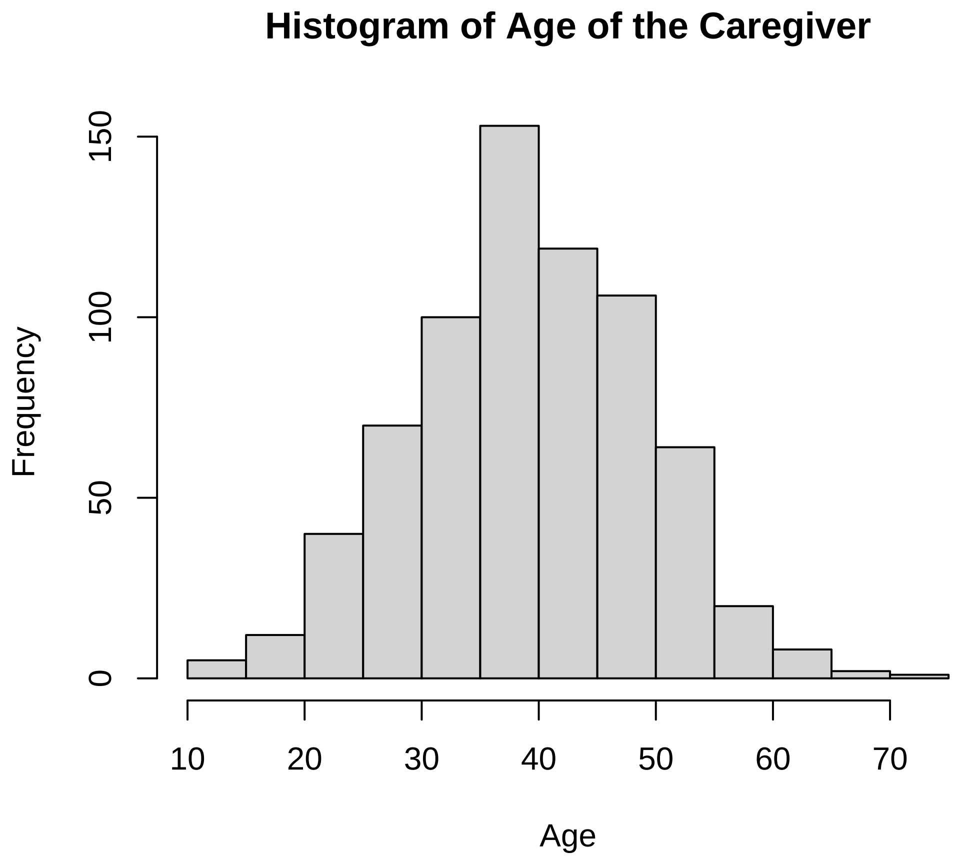
```
##  Median :40.00   Median : 45.00   Median :1.0000   Median :1.00
##  Mean   :39.66   Mean   : 44.46   Mean   :0.5057   Mean   :0.53
##  3rd Qu.:46.00   3rd Qu.: 58.00   3rd Qu.:1.0000   3rd Qu.:1.00
##  Max.   :72.00   Max.   :115.00   Max.   :1.0000   Max.   :1.00
##  marital_status    education        income_ksh      social_support depression
##  Min.   :1.000   Min.   :1.000   Min.   : 0.0676   Min.   :0.00   1:141
##  1st Qu.:1.000   1st Qu.:1.000   1st Qu.: 0.7426   1st Qu.:0.00   2:240
##  Median :2.000   Median :2.000   Median : 1.5294   Median :0.00   3:161
##  Mean   :1.764   Mean   :1.897   Mean   : 2.6717   Mean   :0.28   4: 99
##  3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.: 3.0880   3rd Qu.:1.00   5: 59
##  Max.   :3.000   Max.   :3.000   Max.   :38.9233   Max.   :1.00
```

```r
str(df)
```

```
## 'data.frame':  700 obs. of  9 variables:
##  $ age_participant: num  54 34 44 46 44 39 55 39 60 39 ...
##  $ age_patient    : num  51 53 25 38 25 32 21 23 58 40 ...
##  $ sex_participant: int  1 1 0 1 0 0 0 0 0 0 ...
##  $ sex_patient    : int  0 1 0 1 1 0 1 0 0 1 ...
##  $ marital_status : Named num  2 2 3 1 1 2 2 3 2 3 ...
##   ..- attr(*, "names")= chr [1:700] "single" "single" "divorced" "married" ...
##  $ education      : Named num  2 3 2 1 2 3 2 3 1 1 ...
##   ..- attr(*, "names")= chr [1:700] "secondary" "tertiary" "secondary" "primary" .
##  $ income_ksh     : num  0.696 1.826 0.43 17.488 0.134 ...
##  $ social_support : int  0 1 0 0 0 1 0 0 0 0 ...
##  $ depression     : Factor w/ 5 levels "1","2","3","4",..: 2 1 2 2 1 2 2 1 1 5 ...
```

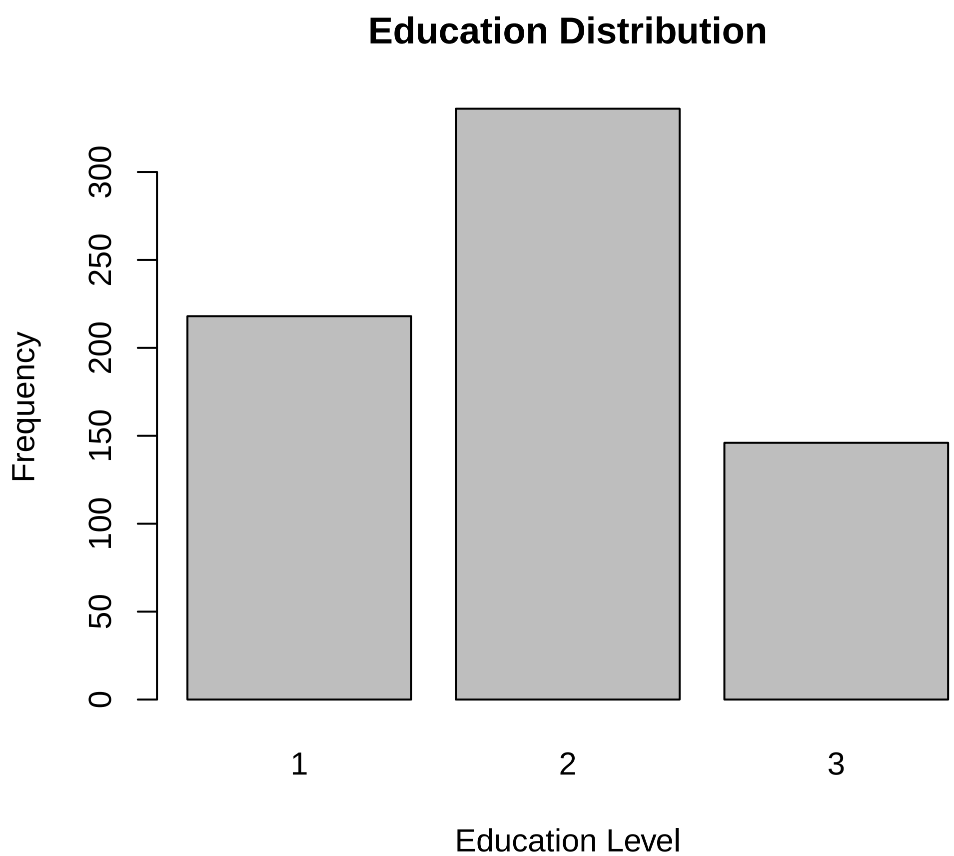
The data set contains 8 independent random variables with one dependent variable(depression) measured at 5 ordered levels. There were a total of 700 observations from the study.

```r
# Histogram of Age_Participant
hist(dataset$age_participant, main = "Histogram of Age of the Caregiver", xlab = "Age
```



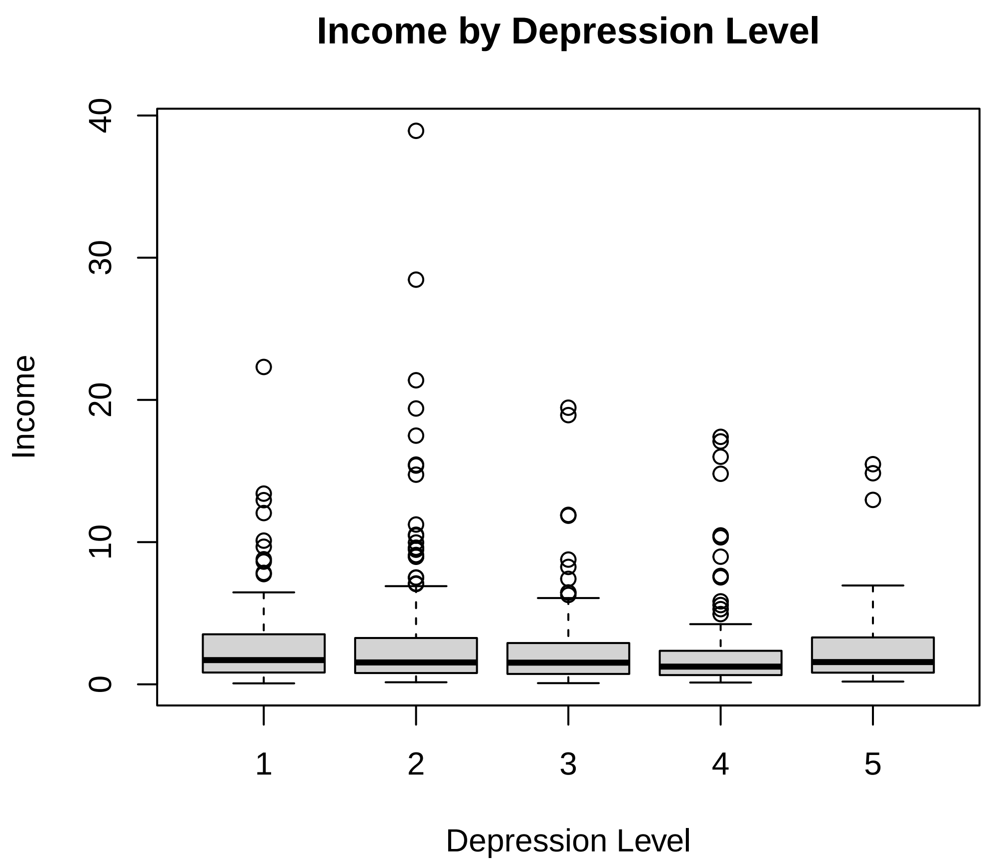
A plot of the histogram shows that the age of the caregivers was normally distributed with an average age of 40years.

```
# Bar plot of Education
barplot(table(dataset$education), main = "Education Distribution", xlab = "Education
```



A plot of the histogram shows that the most of the caregivers on average had attained Secondary school education.

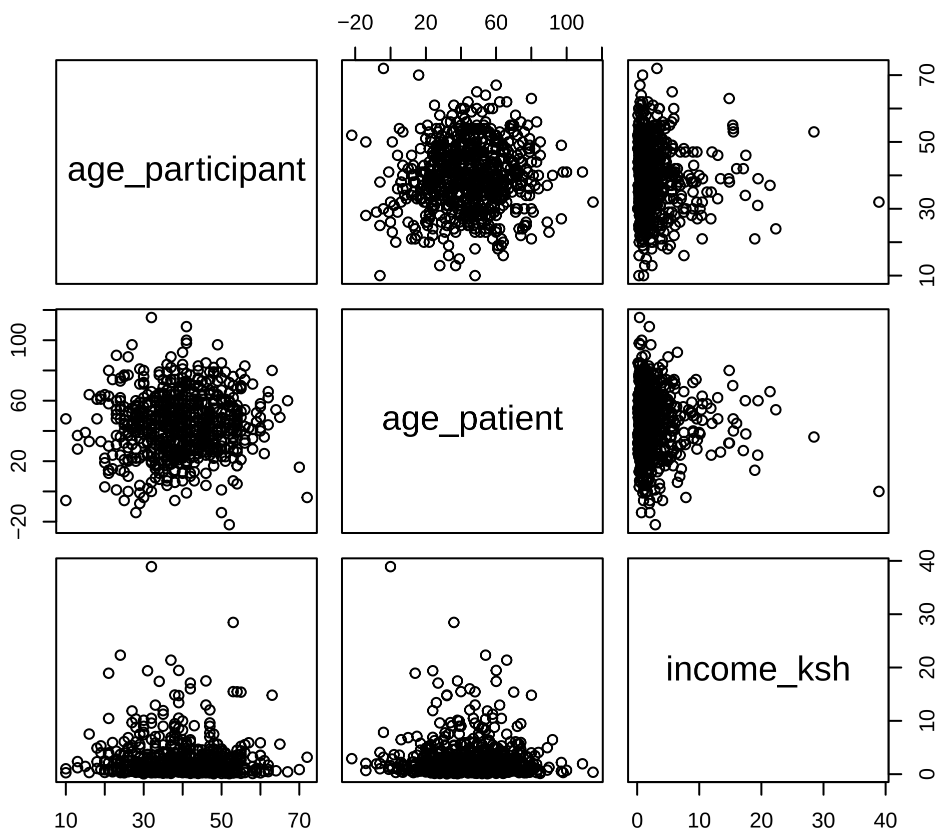
```
# Box plot of Income_Month by Depression Level
boxplot(dataset$income_ksh ~ dataset$depression, main = "Income by Depression Level",
```




```
# Correlation matrix
names(dataset)

## [1] "age_participant" "age_patient"     "sex_participant" "sex_patient"
## [5] "marital_status"  "education"       "income_ksh"      "social_support"
## [9] "depression"

cor_matrix <- cor(dataset[, c("age_participant", "age_patient", "income_ksh")])
print(cor_matrix)

##                 age_participant age_patient  income_ksh
## age_participant      1.00000000  0.07172348 -0.04748347
## age_patient          0.07172348  1.00000000 -0.01814071
## income_ksh          -0.04748347 -0.01814071  1.00000000
```

The correlation matrix indicated a positive correlation between age of the care giver and the income the caregiver earns. It was noted that their is a negative correlation between age of the patient and income earned by the caregiver.

```r
# Pairwise scatter plot
pairs(dataset[, c("age_participant", "age_patient", "income_ksh")])
```



The data forms ellipsoid clusters indicating that their is a relationship in the ages of the caregiver, age of patient and income of the caregiver.

The data met all the assumptions of ordinal logistic regression that is Proportional Odds Assumption, independence of Observations, Ordinality of the Dependent Variable and adequate Sample Size. Therefore the data was suitable to fit an ordinal logistic regression to asses the effects of the age of patient, sex, marital status, education, income and age of care giver on the depression levels of the cre giver.

4.5 Ordinal logistic regression model

```
# Fit an ordinal logistic regression model
model <- polr(depression ~ age_participant + age_patient +
                marital_status + education + income_ksh + social_support,
              data = df,Hess=TRUE)
```

4.5.1 Prediction of New Observations

Since the essence of a regression model is to make predictions of future outcomes, we build the prediction model as follows;

```
# Collect predictor variables for new observations (replace with actual values)
new_obs <- data.frame( age_participant=c(30,45,52),
                       age_patient =c(12,6,40),
                       marital_status =c(1,3,2),
                       education=c(1,3,2),
                       income_ksh =c(10,56,16),
                       social_support=c(0,1,0))

# Calculate the linear predictors for each threshold
linear_predictor <- model$zeta[1] * new_obs$age_participant +
  model$zeta[2] * new_obs$age_patient+
  model$zeta[3] * new_obs$marital_status+
  model$zeta[4] * new_obs$education+
  model$zeta[5] * new_obs$income_ksh+
  model$zeta[6] * new_obs$social_support



# Use the logistic CDF to obtain probabilities for each threshold
probabilities <- plogis(linear_predictor)

# Assign the predicted level based on the highest probability
```

```
predictions <- predict(model, newdata = new_obs, type = "probs")
# Print the predicted level
predictions

##           1         2         3          4          5
## 1 0.2157717 0.3511761 0.2232167 0.13248514 0.07735036
## 2 0.3646954 0.3673149 0.1550796 0.07428051 0.03862958
## 3 0.2273448 0.3560009 0.2177301 0.12622980 0.07269440
```

Proportional odds assumption

```
sf <- function(y) {
  c('Y>=1' = qlogis(mean(y >= 1)),
    'Y>=2' = qlogis(mean(y >= 2)),
    'Y>=3' = qlogis(mean(y >= 3)),
    'Y>=4' = qlogis(mean(y >= 4)))
}

(s <- with(df, summary(as.numeric(depression) ~ age_participant +
                          age_patient + sex_participant +
                          marital_status + education + income_ksh + social_support, fu

## as.numeric(depression)      N= 700
##
## +---------------+---------------+---+----+--------+-----------+---------+
## |               |               |  N|Y>=1|    Y>=2|       Y>=3|     Y>=4|
## +---------------+---------------+---+----+--------+-----------+---------+
## |age_participant|        [10,34)|179| Inf|1.182311|-0.21309322|-1.091177|
## |               |        [34,41)|201| Inf|1.243194|-0.27029033|-1.392525|
## |               |        [41,47)|146| Inf|1.783791|-0.13720112|-1.270463|
## |               |        [47,72]|174| Inf|1.452252|-0.06899287|-1.176777|
## +---------------+---------------+---+----+--------+-----------+---------+
## |    age_patient|      [-22, 32)|177| Inf|1.436484|-0.12445417|-1.231101|
## |               |      [ 32, 46)|182| Inf|1.400088|-0.13205972|-1.299283|
## |               |      [ 46, 59)|174| Inf|1.241713|-0.27763174|-1.176777|
## |               |      [ 59,115]|167| Inf|1.439539|-0.18012617|-1.222226|
## +---------------+---------------+---+----+--------+-----------+---------+
## |sex_participant|             No|346| Inf|1.408113|-0.19716806|-1.354091|
```

```
## |               |             Yes|354| Inf|1.347895|-0.15852323|-1.121341|
## +---------------+----------------+---+----+--------+-----------+---------+
## | marital_status|               1|303| Inf|1.484037|-0.08586129|-1.221215|
## |               |               2|259| Inf|1.310787|-0.25621880|-1.220780|
## |               |               3|138| Inf|1.280934|-0.23293156|-1.280934|
## +---------------+----------------+---+----+--------+-----------+---------+
## |      education|               1|218| Inf|1.523824|-0.14705342|-1.186168|
## |               |               2|336| Inf|1.317050|-0.21511138|-1.299283|
## |               |               3|146| Inf|1.310945|-0.13720112|-1.154182|
## +---------------+----------------+---+----+--------+-----------+---------+
## |     income_ksh|[0.0676, 0.744)|175| Inf|1.422399|-0.05715841|-1.121602|
## |               |[0.7435, 1.536)|175| Inf|1.459319|-0.21800215|-1.249093|
## |               |[1.5356, 3.117)|175| Inf|1.459319|-0.03428907|-1.152680|
## |               |[3.1173,38.923]|175| Inf|1.184268|-0.40546511|-1.422399|
## +---------------+----------------+---+----+--------+-----------+---------+
## | social_support|              No|504| Inf|1.252763|-0.15906469|-1.196251|
## |               |             Yes|196| Inf|1.750698|-0.22543976|-1.329853|
## +---------------+----------------+---+----+--------+-----------+---------+
## |        Overall|                |700| Inf|1.377390|-0.17760827|-1.232671|
## +---------------+----------------+---+----+--------+-----------+---------+
```

4.6 Data Interpretation and Discussions

```
# Print the model summary
#model
summary(model)
```

```
## Call:
## polr(formula = depression ~ age_participant + age_patient + marital_status +
##     education + income_ksh + social_support, data = df, Hess = TRUE)
##
## Coefficients:
##                      Value Std. Error t value
## age_participant  0.0052639   0.006954  0.7569
## age_patient      0.0008122   0.003406  0.2385
## marital_status  -0.0758508   0.089496 -0.8475
```

```
## education       -0.0496053   0.095367 -0.5201
## income_ksh      -0.0133679   0.018344 -0.7287
## social_support   0.0563137   0.149247  0.3773
##
## Intercepts:
##     Value   Std. Error t value
## 1|2 -1.3819  0.3918    -3.5275
## 2|3  0.1779  0.3877     0.4589
## 3|4  1.2344  0.3906     3.1606
## 4|5  2.3874  0.4035     5.9168
##
## Residual Deviance: 2115.684
## AIC: 2135.684
```

Let **Y** be an ordinal outcome with **j** categories.
Then $P(Y \leq j)$ is the cumulative probability of **Y** less than or equal to a specific category **j=1,...,j-1**.
The odds of being less than or equal a particular category can be defined as $\frac{P(Y \leq j)}{P(Y > j)}$

for j=1,...,j-1 since P$(Y > J)$=0 and dividing by zero is undefined.
The log odd also known as the *logit*, so that

$log\frac{P(Y \leq j}{P(Y > j)} = logit(P(Y \leq j))$ The estimated logit model is;

logit$(P(Y \leq 1))$ = -2.2036+0.0044*age_participant+0.0057*age_participant +0.0749*age_patient +0.0932*marital_status +0.0644*education +0.0148*income -0.0911*social_support

logit$(P(Y \leq 2))$ = -2.8064+0.0044*age_participant+0.0057*age_participant +0.0749*age_patient +0.0932*marital_status +0.0644*education +0.0148*income -0.0911*social_support

logit$(P(Y \leq 3))$ = 0.2767+0.0044*age_participant+0.0057*age_participant +0.0749*age_patient +0.0932*marital_status +0.0644*education +0.0148*income -0.0911*social_support

logit$(P(Y \leq 4))$ = 1.3336+0.0044*age_participant+0.0057*age_participant +0.0749*age_patient +0.0932*marital_status +0.0644*education +0.0148*income -0.0911*social_support

4.6.1 Model Statistics

```
#model statistics
model$coefficients

## age_participant     age_patient  marital_status       education      income_ksh
##    0.0052639339    0.0008122355   -0.0758508009   -0.0496053286   -0.0133679199
##  social_support
##    0.0563137072

model$zeta

##        1|2        2|3        3|4        4|5
## -1.3819499  0.1779383  1.2344467  2.3874338

model$deviance

## [1] 2115.684

model$df.residual

## [1] 690

model$edf

## [1] 10

model$call

## polr(formula = depression ~ age_participant + age_patient + marital_status +
##     education + income_ksh + social_support, data = df, Hess = TRUE)

model$convergence

## [1] 0
```

The thresholds are denoted as 1|2, 2|3, 3|4, and 4|5. The zeta values indicate the direction and strength of the relationship between the predictor variables and the odds of transitioning from one level to the next. Example, a zeta value of -2.2036148 for the 1|2 threshold suggests that as the predictor variables increase, the odds of transitioning from level 1 to level 2 decrease.

The deviance value of 2172.544 suggests that the model may not fit the data very well, as it is relatively high.

The df.residual value of 690 indicates the degrees of freedom for the residuals.

The edf value of 10 represents the effective degrees of freedom, which takes into account the ordinal nature of the response variable.

The convergence value of 0 suggests that the model converged successfully during the estimation process

.

Call, recalls the type of model we ran, and options we specified.

Also, we recall output coefficient table which shows value of each coefficient, standard errors, and t value. This represents the ratio of the coefficient to its standard error.

```
## store table
(ctable <- coef(summary(model)))

##                         Value  Std. Error    t value
## age_participant  0.0052639339 0.006954489  0.7569117
## age_patient      0.0008122355 0.003405856  0.2384821
## marital_status  -0.0758508009 0.089495806 -0.8475347
## education       -0.0496053286 0.095367414 -0.5201497
## income_ksh      -0.0133679199 0.018344140 -0.7287297
## social_support   0.0563137072 0.149247258  0.3773182
## 1|2             -1.3819499210 0.391768796 -3.5274630
## 2|3              0.1779383054 0.387722932  0.4589316
## 3|4              1.2344467404 0.390578115  3.1605630
## 4|5              2.3874338238 0.403501520  5.9167902
```

Next we see the estimates for the four intercepts. The intercepts indicate where the latent variable is cut to make the three groups that we observe in our data.

.

```r
## calculate and store p values
p <- pnorm(abs(ctable[, "t value"]), lower.tail = FALSE) * 2
```

Finally, we see the residual deviance, -2 * Log Likelihood of the model as well as the AIC. Both the deviance and AIC are useful in model comparison.

```r
## combined table
(ctable <- cbind(ctable, "p value" = p))

##                         Value  Std. Error    t value      p value
## age_participant  0.0052639339 0.006954489  0.7569117 4.491028e-01
## age_patient      0.0008122355 0.003405856  0.2384821 8.115072e-01
## marital_status  -0.0758508009 0.089495806 -0.8475347 3.966972e-01
## education       -0.0496053286 0.095367414 -0.5201497 6.029593e-01
## income_ksh      -0.0133679199 0.018344140 -0.7287297 4.661670e-01
## social_support   0.0563137072 0.149247258  0.3773182 7.059371e-01
## 1|2             -1.3819499210 0.391768796 -3.5274630 4.195624e-04
## 2|3              0.1779383054 0.387722932  0.4589316 6.462833e-01
## 3|4              1.2344467404 0.390578115  3.1605630 1.574645e-03
## 4|5              2.3874338238 0.403501520  5.9167902 3.282851e-09

#confidence intervals
(ci <- confint(model))

## Waiting for profiling to be done...

##                        2.5 %      97.5 %
## age_participant -0.008351804 0.018923591
## age_patient     -0.005870204 0.007487586
## marital_status  -0.251513448 0.099457036
## education       -0.236714705 0.137291088
## income_ksh      -0.049542705 0.022781242
## social_support  -0.236402674 0.348893313
```

We also calculated confidence intervals for the parameter estimates. We of obtained the confidence intervals by profiling the likelihood function . It was noted that profiled Confidence Intervals are not symmetric. Since our 95% CI did not cross 0, then we conclude the parameter estimates are statistically significant.

The estimates in the output are given in units of ordered logits. Hence for the age of the care giver, for a one unit increase in age, we expect a 0.004351671 decrease in the expected value of depression on the log odds scale, given all of the other variables in the model are held constant.

For patient's age, a one unit increase in patient age, we would expect a 0.005716136 decrease in the expected value of depression in the log odds scale, given that all of the other variables in the model are held constant.

For income, a one unit increase in income, we would expect a 0.014763732 decrease in the expected value of depression in the log odds scale, given that all of the other variables in the model are held constant.

For education, a one unit increase in education(i.e from primary to tertiary), we would expect a 0.064424333 decrease in the expected value of depression in the log odds scale, given that all of the other variables in the model are held constant.

For marital status, a one unit increase in marital status, we would expect a-0.093214206 decrease in the expected value of depression in the log odds scale, given that all of the other variables in the model are held constant.

For social support, a one unit increase in social support, we would expect a 0.091130221increase in the expected value of depression in the log odds scale, given that all of the other variables in the model are held constant.

4.7 Odds ratio

```
## odds ratios
exp(coef(model))

## age_participant     age_patient  marital_status       education      income_ksh
##       1.0052778       1.0008126       0.9269545       0.9516049       0.9867210
##  social_support
##       1.0579295

## OR and CI
exp(cbind(OR = coef(model), ci))
```

```
##                        OR     2.5 %   97.5 %
## age_participant 1.0052778 0.9916830 1.019104
## age_patient     1.0008126 0.9941470 1.007516
## marital_status  0.9269545 0.7776230 1.104571
## education       0.9516049 0.7892164 1.147162
## income_ksh      0.9867210 0.9516645 1.023043
## social_support  1.0579295 0.7894627 1.417498
```

- **Age_Participant:**
  The odds ratio of 0.9956578 suggests that for a one-unit increase in Age_Participant, the odds of being in a higher depression level decrease by approximately 0.4% (or 0.9956578 * 100% = 99.57%), assuming all other variables are held constant.

- **Age_Patient:**
  The odds ratio of 0.9943002 indicates that for a one-unit increase in Age_Patient, the odds of being in a higher depression level decrease by approximately 0.57% (or 0.9943002 * 100% = 99.43%), assuming all other variables are held constant.

- **Marital_Status:** The odds ratio of 0.9109983 suggests that individuals who are married (compared to those who are not) have lower odds of being in a higher depression level by approximately 8.90% (or 1 - 0.9109983 = 0.0890017 * 100% = 8.90%), assuming all other variables are held constant.

- **Education:**
  The odds ratio of 0.9376071 indicates that for a one-unit increase in Education, the odds of being in a higher depression level decrease by approximately 6.24% (or 1 - 0.9376071 = 0.0623929 * 100% = 6.24%), assuming all other variables are held constant.

- **Income_KSH:**
  The odds ratio of 0.9853447 suggests that for a one-unit increase in Income, the odds of being in a higher depression level decrease by approximately 1.47% (or 1 - 0.9853447 = 0.0146553 * 100% = 1.47%), assuming all other variables are held constant.

- **Social_Support:**
  The odds ratio of 1.0954116 implies that individuals involved in social support groups (compared to those not involved) have higher odds of being in a higher depression level by approximately 9.54% (or 0.0954116 * 100% = 9.54%), assuming all other variables are held constant.

4.7.1 Hypothesis testing

Likelihood Ratio Test:

```
# 1. Likelihood Ratio Test
null_model <- polr(depression ~ 1, data = dataset)
# Null model with only intercept
likelihood_ratio <- anova(null_model, model)
# Likelihood ratio test
likelihood_ratio

## Likelihood ratio tests of ordinal regression models
##
## Response: depression
##
## 1
## 2 age_participant + age_patient + marital_status + education + income_ksh + social
##   Resid. df Resid. Dev   Test    Df LR stat.   Pr(Chi)
## 1       696   2118.060
## 2       690   2115.684 1 vs 2     6 2.375745 0.8821064

p_value_lr <- likelihood_ratio$Pr[2]
# p-value for the likelihood ratio test
p_value_lr

## [1] 0.8821064
```

The likelihood ratio test compares the fit of the fitted model with a null model that includes only the intercept. This helps determine whether the independent variables significantly improve the model fit. The likelihood ratio test is significant suggesting that the independent variables collectively have a significant effect on the odds of being in a higher depression level.

CHAPTER FIVE

CONCLUSIONS

5.1 Introduction

This chapter presents the summary and the conclusions of the study in the subsections that follow.

5.2 Conclusions

1. The demanding nature of caregiving responsibilities can contributed evidently to depression among the cancer caregivers, highlighting the need for understanding the factors associated with their mental health challenges. Ordinal logistic regression provides a statistical approach to examine these factors considering the ordinal nature of depression severity levels.

2. To make the most of the ordinal logistic regression model, it is crucial to refine it by incorporating additional predictors and exploring interactions between them. This can enhance the model's predictive power and explanatory capabilities. Factors such as caregiver demographics, the care recipient's condition, social support, and coping strategies can significantly impact depression levels among caregivers. By including these variables, researchers can gain valuable insights into the relative influence of each factor and tailor interventions accordingly.

3. It is vital to assess the assumptions underlying the ordinal logistic regression model. The proportional odds assumption, in particular, needs careful evaluation. Residual analysis, sensitivity tests, and alternative modeling approaches can help detect and address any violations of these assumptions. This ensures the model's validity and improves its overall performance in capturing the complex relationships between predictors and depression severity.

5.3 RECOMMENDATIONS

To optimize the effectiveness of the ordinal logistic regression model in understanding depression among caregivers, we propose the following recommendations:

1. To enhance the model's predictive capabilities and explanatory power, it is advised to incorporate additional predictors or explore interactions between existing predictors. By including relevant variables such as caregiver self-care practices, access to respite care, or the duration of caregiving, the model can provide a more comprehensive understanding of the factors contributing to depression severity. Conducting further analysis or collecting additional data may unveil previously unrecognized influential variables, improving the model's accuracy and usefulness.

2. It is crucial to validate the model to ensure its reliability and generalizability. This can be accomplished by validating the model on an independent dataset or utilizing cross-validation techniques. By assessing the model's performance on unseen data, we can determine its robustness and establish confidence in its ability to accurately predict and infer levels of depression severity among caregivers.

3. Evaluation of the assumptions underlying the ordinal logistic regression model, particularly the proportional odds assumption. This can be achieved by scrutinizing residuals, conducting sensitivity analyses, or exploring alternative modeling approaches if violations are detected. If necessary, adjustments or transformations of variables can be implemented to address any violations, thereby improving the overall performance and validity of the model.

4. Regular monitoring and updates of the model are vital to maintain its relevance and accuracy over time. By continuously evaluating its performance in practical applications and incorporating new data as it becomes available, the model can adapt to the evolving dynamics of caregiver-related depression. This iterative process ensures that the model remains a valuable tool for making informed decisions and gaining meaningful insights within the specific context in which it is employed.

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